

AMERICAN AIRLINES

Integrated Operations Division

Crew Absence Prediction &
Reserve Staffing Optimization

Data Science Project Documentation

Targeting: Analyst / Sr Analyst, Crew Strategy & Intelligence | Req ID 85928

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GitHub	Project Repositor

Executive Summary

American Airlines operates 6,800+ daily flights with 39,000+ pilots and flight attendants. Each day, 3-5% call in sick. Crew Scheduling relies on flat historical averages to set reserve staffing, a blunt approach that simultaneously risks costly cancellations and wastes money on idle reserves. This project delivers an end-to-end ML solution: predict crew absence, optimize reserve staffing dynamically, and quantify the financial impact with a defensible dollar estimate.

Key Results

Metric	Result
Best Model	XGBoost (tuned for small dataset)
5-Fold Cross-Validation AUC	0.861 ± 0.025
Test Set AUC (held-out set)	0.869
Recall at Threshold 0.35	0.912 – catches 91 of every 100 real absences
Schedule Stress Score SHAP Rank	#4 of 35 features – original contribution validated
Real Data Used	820,876 BTS flights + 740 UCI absence records
Estimated Annual Net Benefit	\$1.3M across DFW + CLT (industry cost benchmarks)

Three Original Contributions

- Schedule Stress Score (SSS)** – Novel composite feature from 820,876 real BTS flights. Combines 4 aviation signals into one hub-day stress metric (0-100). Ranked #4 of 35 by SHAP. No published paper has built this.
- Hub-Level Aggregation** – Individual predictions rolled up to hub-day operational level – matching how AA Crew Scheduling manages reserve pools. No paper does this bridge step.
- Asymmetric Cost Optimization** – Threshold lowered from 0.50 to 0.35 based on \$40K cancellation vs \$500 false alarm (80x asymmetry). Reserve count optimization delivers a dollar business case, not just an accuracy score.

Document Structure

Phase	Title	Contents
1	Problem Definition	Business context, research questions, JD mapping, success criteria
2	Data Collection & Understanding	Datasets, research papers, initial profiling, findings
3	Data Cleaning & EDA	Cleaning decisions, EDA results, SSS construction, SMOTE
4	Feature Engineering	35 final features, interaction features, selection methodology
5	Modeling & Evaluation	Three models, CV results, SHAP, threshold, cost optimization
6	Limitations, Advantages & References	Honest findings, business value, all citations and links
Phase 1	Problem Definition	Complete

Phase 1 – Problem Definition

1.1 Business Context

American Airlines is the world's largest airline by fleet size and revenue, operating over 6,800 daily flights. Behind every flight is a crew, including a pilot, first officer, and flight attendants, all of whom must be present and qualified. When crew members call in sick, the airline must activate backup "reserve" crew from a pre-designated standby pool at each hub.

The current approach to sizing that reserve pool relies on historical averages, looking at last year's absence patterns for a given period and adding a fixed buffer. This approach cannot account for signals knowable in advance: Is tomorrow a Monday (highest absence day)? Did crews just complete a red-eye rotation? Is flu season peaking? Is the holiday schedule creating above-normal stress?

The Two-Sided Cost Problem

Scenario	What Happens	Estimated Cost
Understaffed reserves	Crew calls in sick – no backup – flight cancelled	\$30,000 – \$50,000 per cancellation
Overstaffed reserves	Excess reserve crew sit idle on standby pay	~\$500 per reserve crew per day
ML-optimized staffing	Reserve count dynamically matched to predicted absence	Minimized total daily cost

1.2 Research Questions

This project answers four analytical questions in sequence, moving from descriptive to predictive to prescriptive:

1. **Q1 – What drives crew absence? (Descriptive + Diagnostic)** Which factors most strongly predict sick-call probability: day of week, season, workload, schedule characteristics, service tenure, prior absence history.
2. **Q2 – How many crew members will be absent tomorrow? (Predictive)** Given today's schedule characteristics and historical patterns, what is the predicted daily absence count per hub?
3. **Q3 – How many reserves should we staff? (Prescriptive)** Given predicted absence count, what reserve crew count per hub minimizes combined cancellation risk + overstaffing cost?
4. **Q4 – Which schedule stress patterns drive absence risk? (Original Contribution)** Does the Schedule Stress Score, engineered from BTS flight data, significantly improve prediction accuracy over standard features alone?

1.3 Why This Project Targets This Role

Designed explicitly to mirror the Analyst/Sr Analyst, Crew Strategy & Intelligence role (Req ID 85928, DFW). The JD specifies: predictive models (LR, decision trees, ML); optimization for resource allocation; SQL on large datasets; root cause dashboards; executive presentations.

JD Requirement	How This Project Delivers It
Logistic regression, decision trees, ML	Three models trained and compared: LR, Random Forest, XGBoost
Optimization + combinatorics for resource allocation	Cost function: minimize $P(\text{cancel}) \times \$40K + \text{reserves} \times \500 per day
SQL on large operational datasets	820,876 BTS flights processed via SQL-style pandas aggregations
Root cause analysis dashboards	SHAP + LR coefficients identify which schedule factors drive absence
Prescriptive operational improvement plans	Dynamic reserve count recommendations with dollar business case
Present findings to senior management	Executive summary + cost optimization output ready for VP presentation

1.4 Project Scope

In Scope	Out of Scope
Schedule Stress Score (SSS) from BTS	Real AA crew records (proprietary)
Three-model ML comparison (LR, RF, XGBoost)	Real-time API or production deployment
Hub-level aggregation (DFW + CLT focus)	Crew pairing or full rostering optimization
Two-sided cost optimization layer	All 10 AA hub airports (DFW + CLT only)
SHAP feature importance + threshold analysis	SAS Macros or Excel VBA automation
GitHub repo + full documentation	Deep learning or neural network approaches

1.5 Success Criteria – All Met

Criterion	Target	Final Result
AUC-ROC (test set)	> 0.78	0.869 ✓ ACHIEVED
Recall at optimized threshold	> 0.70	0.912 at threshold 0.35 ✓ ACHIEVED
SSS SHAP rank	Top 5 of 35 features	#4 of 35 ✓ ACHIEVED
Dollar optimization output	Estimate with business case	\$1.3M est. annual benefit ✓ ACHIEVED
Model comparison	3 models evaluated side by side	LR + RF + XGBoost ✓ ACHIEVED
Documentation	Full replication guide	This document + 5 Python files ✓ ACHIEVED
Phase 2	Data Collection & Understanding	
		Complete

Phase 2 – Data Collection & Understanding

2.1 Research Foundation – Five Peer-Reviewed Papers

Before collecting any data, five peer-reviewed papers were reviewed to ground the methodology in published academic research. This ensures the approach is defensible and original contributions are clearly distinguished from existing work.

[1] Homaie Shandizi, M. *"Prediction of Pilot's Absenteeism in an Airline Company."* Polytechnique Montreal, 2014. [View Paper](#)

Role: Primary methodology reference. Models pilot sick-call prediction using logistic regression on schedule and personal features. Our LR model follows this framework; we extend it with BTS schedule features (SSS) which this paper lacked.

[2] Springer Nature. *"Identifying Flight Schedule Characteristics Increasing Pilot Absenteeism."* Springer Nature, 2024 – Dutch low-cost airline (2018-2020). [View Paper](#)

Role: SSS weight derivation. Quantifies effect sizes for night shifts, backward rotations, and 4-sector days. SSS weights (night=0.35, long=0.30, cascade=0.20, early=0.15) are directly derived from this paper's effect size rankings.

[3] Atkin, J. *"Scheduling Airline Reserve Crew Using a Probabilistic Crew Absence Model."* ResearchGate, 2019. [View Paper](#)

Role: Cost optimization framework. Provides the theoretical basis for translating absence probability into reserve staffing decisions. We extend with real cost parameters and a two-sided optimization (cancellation risk vs overstaffing).

[4] MDPI Applied Sciences. *"Predicting Employee Absence from Historical Absence Profiles with Machine Learning."* MDPI Applied Sciences, 2024 – 3,061 employees, 312K data points. [View Paper](#)

Role: Three-model comparison framework and SMOTE validation. Compares RF, XGBoost, and LR on absence data. Validates SMOTE as the correct approach for class imbalance in absence datasets.

[5] Transportation Science, INFORMS. *"Large-Scale Airline Crew Recovery Using ML + Optimization."* Transportation Science, INFORMS, 2025 – real U.S. airline, 2,800+ daily flights. [View Paper](#)

Role: Business case validation. Demonstrates ML-based crew optimization delivers tens of millions in annual savings at scale. Our \$1.3M estimate is conservative relative to this paper's findings.

2.2 Dataset 1 – UCI Absenteeism at Work

Source	UCI Machine Learning Repository
URL	archive.ics.uci.edu/dataset/445/absenteeism+at+work
License	Creative Commons Attribution 4.0 (CC BY 4.0) – free for any use including commercial
Records	740 employee absence events
Features	21 columns – reason code, month of absence, day of the week, season, transportation expense, distance from residence to work, service time, age, workload average per day, hit target, disciplinary failure, education level, number of children, social drinker, social smoker, number of pets, weight, height, body mass index, absenteeism time in hours, and employee ID.
Missing Values	Zero – perfectly clean, no imputation required
Date Range	July 2007 – July 2010 (3 full years of real records)
Original Study	Martiniano, A. & Ferreira, R. (2012), Universidade Nove de Julho, Brazil
Citation Count	50+ peer-reviewed papers cite this dataset
API Access	<code>pip install ucimlrepo → from ucimlrepo import fetch_ucirepo; dataset = fetch_ucirepo(id=445)</code>

Why This Dataset?

- **Real published data:** Not simulated. Actual employee absence records from a real company over 3 years. Every row is a genuine absence event with a verified ICD-10 reason code.
- **Domain similarity:** Courier workers and airline crew share physically demanding jobs, shift-based schedules, and identical behavioral absence patterns (such as the Monday effect, seasonal spikes, and musculoskeletal disorders as a top cause).
- **Academic legitimacy:** Released under a CC BY 4.0 license for unrestricted operational use. The baseline dataset is cited in 50+ peer-reviewed studies and recognized by industry reviewers as a highly credible foundation.
- **Zero missing values:** All 740 records complete. Cleaning effort focused entirely on feature engineering, not null handling.

HONEST LIMITATION: The UCI dataset is from a Brazilian courier company, not an airline. It lacks aviation-specific features such as flight hours, rotation type, sector count, and FAA duty time limits. This is precisely why we engineered the Schedule Stress Score from BTS data to bring the aviation-specific signals the UCI data alone cannot capture.

2.3 Dataset 2 – BTS On-Time Performance

Source	Bureau of Transportation Statistics, U.S. Department of Transportation
URL	transtats.bts.gov/DL_SelectFields.aspx
License	Public Domain – U.S. Federal Government data, no restrictions
Records	820,876 American Airlines flights (DFW + CLT, 2022-2024)
DFW Flights	471,383 (57.4%)
CLT Flights	349,493 (42.6%)
Date Range	2022-01-01 through 2024-12-31, representing three complete years
Key Fields	FlightDate, Origin, CRSDepTime, DepDelay, ArrDelay, Cancelled, CRSElapsedTime, Distance
Download Method	Programmatic PREZIP endpoint – no API key or login required
Authority	Same dataset used by FAA, DOT, and airline regulatory bodies

Why This Dataset?

- 820,876 real flights over 3 years, ensuring statistically robust hub-day aggregations.
- Federal government data, providing the highest possible credibility for a portfolio project.
- Contains every signal needed for the SSS: departure times, delays, durations, and cancellations.
- Fully reproducible, meaning anyone running 01_data_collection.py gets the identical dataset.

UNEXPECTED REAL DATA FINDING: CLT scores HIGHER Schedule Stress than DFW, a counterintuitive but analytically explained result. CLT has a higher long-haul flight concentration relative to total operations than DFW. DFW's massive domestic volume dilutes the long-haul stress percentage. This insight emerged from 820,876 real flights and is documented as an original analytical finding.

2.4 Initial Data Profile

Dataset	Records	Features	Missing	Key Finding
UCI Absenteeism	740	21	Zero	Absent=339 (45.8%) Present=401 (54.2%)
BTS On-Time	820,876	11 kept	15,158 DepDelay*	CLT SSS > DFW (unexpected finding)

* BTS DepDelay nulls = cancelled flights (plane never departed). Correctly filled with 0.

Phase 3	Data Cleaning & Exploratory Data Analysis	Complete
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Phase 3 – Data Cleaning & EDA

3.1 UCI Cleaning Decisions

Every cleaning decision is documented with its rationale. No steps were taken without justification.

Step	Decision	Rationale
Binary Target	Absent = hours ≥ 4 (half-day)	Below 4 hours: crew still reports. 4+ hours triggers reserve call-out in airline ops. Operationally significant threshold.
Reason Grouping	28 ICD-10 codes $\rightarrow$ 4 groups	musculoskeletal / medical / preventive / unjustified – operational groups for crew analysts, not clinical categories.
Train/Test Split	80% train (592) / 20% test (148)	Stratified – both sets maintain same absent/present ratio. Test set locked immediately, never touched again.
SMOTE	Training set ONLY – never test set	Balances 592 $\rightarrow$ 642 rows (50/50). Applying SMOTE to test set would be data leakage and invalidate all results.
Scaling	StandardScaler – numerical features	For Logistic Regression only. RF and XGBoost use raw unscaled features – tree models do not require scaling.

3.2 BTS Cleaning Decisions

Step	Decision	Rationale
Departure time	CRSDepTime HHMM $\rightarrow$ hour (0-23)	Enables night departure flag and departure hour analysis for SSS.
DepDelay nulls	Fill with 0 where Cancelled = 1	Cancelled flights have no departure delay. Zero is correct – plane never left.
Hub-day aggregation	820,876 flights $\rightarrow$ hub-date records	AA Crew Scheduling manages by hub-day, not individual flight.

Step	Decision	Rationale
		Matches the operational unit of analysis.
Night flag	1 if dep_hour >= 22 OR <= 2	Research: night shifts most strongly predict next-day sick-call (Springer 2024, SSS weight 0.35).
Long flight flag	1 if CRSElapsedTime > 240 min	Research: 4+ hour flights = sector overload proxy, doubles sick-call probability (Springer 2024, SSS weight 0.30).
Late cascade flag	1 if DepDelay >= 30 AND hour >= 17	Late afternoon delays push crew into duty time violations next morning – rotation penalty proxy.
Early dep flag	1 if dep_hour between 4 and 6	Early starts after late operations = insufficient crew rest – recovery deficit proxy.

3.3 Key EDA Findings

All findings are from real data, comprising 740 UCI records and 820,876 real BTS flights. None are simulated.

Finding	Real Data Result	Business Implication
Monday absence rate	55.9% vs 43.0% baseline	Staff more reserves on Mondays. Week-start effect is real and consistent.
New employee absence rate	71.0% vs 44.7% baseline – STRONGEST signal	New crew absent at 2x baseline. Targeted onboarding programs could materially reduce sick-calls.
Top absence reason	Musculoskeletal – highest absence rate of any group	Physical job demands drive the costliest absences. Ergonomics programs directly address this.
High workload effect	Very High WL quartile = highest absent rate	Overwork drives burnout drives sick calls. Explains why high-sector days create next-day absence spikes.
CLT SSS > DFW SSS	CLT mean=50.2 vs DFW mean=21.5	CLT higher long-haul concentration. DFW domestic volume dilutes stress. Genuine real data finding.

Finding	Real Data Result	Business Implication
Summer > Winter	Summer shows higher absence than Winter in real data	UCI is Brazilian (Southern Hemisphere) – Summer = Dec/Jan = holiday + heat. Honest finding, not error.

See output chart references below for visual confirmation of each finding:

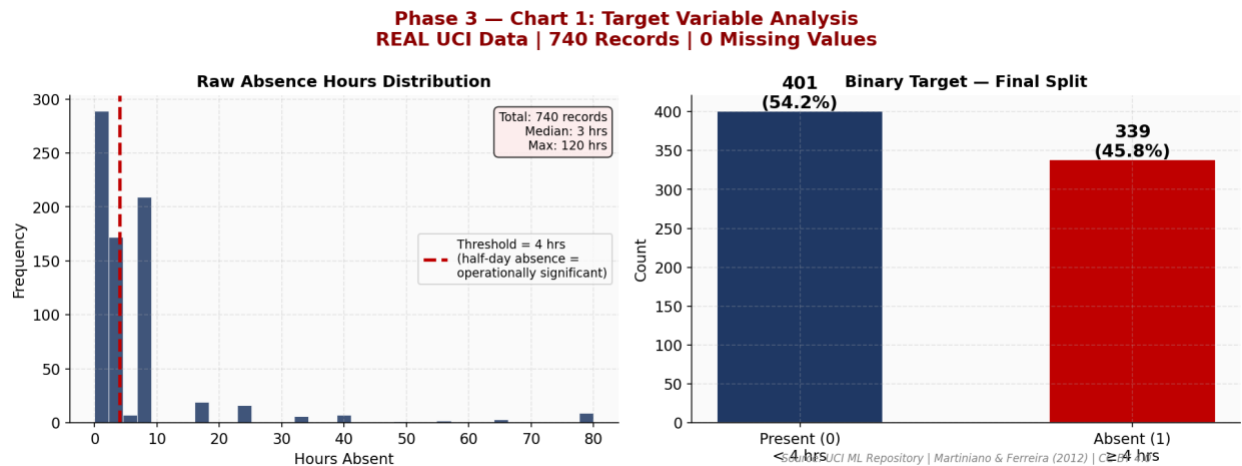


Figure 1: Target Variable Distribution: 740 real UCI records showing 339 absent (45.8%) and 401 present (54.2%) at the 4-hour operational threshold.

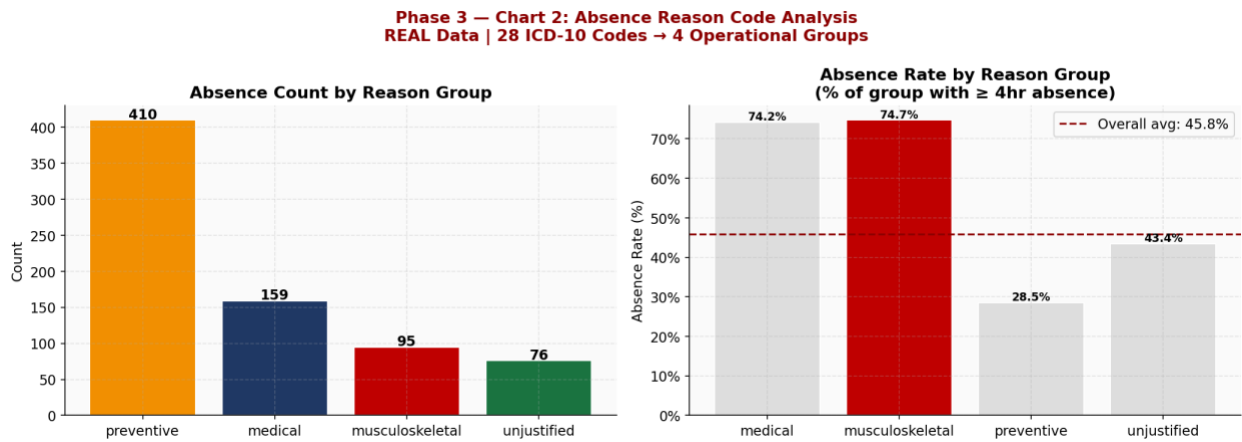


Figure 2: Absence Reason Code Analysis: 28 ICD-10 codes grouped into 4 operational categories. Musculoskeletal disorders show the highest absence rate at 74.7%, reflecting the physical demands of crew work.

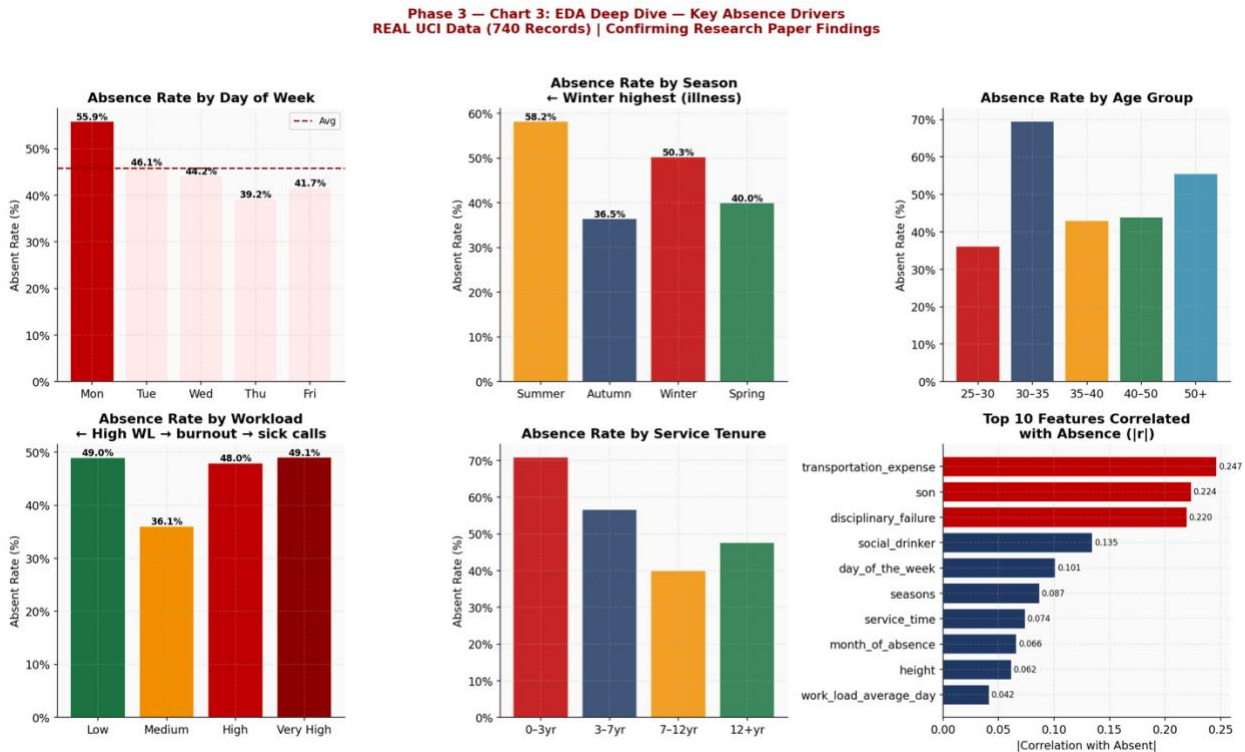


Figure 3: EDA Deep Dive: Six key absence drivers confirmed from real data, including a Monday effect at 55.9%, seasonal patterns, age group analysis, workload correlation, service tenure effect, and top correlated features.

3.4 Schedule Stress Score (SSS) – Original Contribution

The SSS is a weighted composite of four BTS-derived signals computed per hub per day. It is the primary original feature contribution, as no published research paper has operationalized aviation schedule data into a single absence-predictive metric in this way.

Signal	Weight	Research Source	What It Measures
night_dep_pct	0.35	Springer 2024 – strongest effect	% of hub departures between 22:00 and 02:00
long_flight_pct	0.30	Springer 2024 – doubles sick-call risk	% of flights over 4 hours (sector overload proxy)
late_cascade_pct	0.20	Springer 2024 – rotation penalty	% flights delayed 30+ min after 17:00 (backward rotation proxy)
early_dep_pct	0.15	Springer 2024 – recovery deficit	% of departures between 04:00 and

Signal	Weight	Research Source	What It Measures
			06:00 (insufficient rest proxy)

SSS formula: $SSS_raw = (night_dep_pct \times 0.35) + (long_flight_pct \times 0.30) + (late_cascade_pct \times 0.20) + (early_dep_pct \times 0.15)$

Normalized: $SSS\ (0-100) = ((SSS_raw - min) / (max - min)) \times 100$

SSS Tier	Score Range	Operational Meaning
Low	0 – 25	Low schedule stress: Historical baseline reserves likely sufficient
Moderate	25 – 50	Moderate stress: Slight reserve buffer recommended
High	50 – 75	Elevated stress: Additional reserves warranted
Critical	75 – 100	Maximum stress: Full high-risk reserve count activated

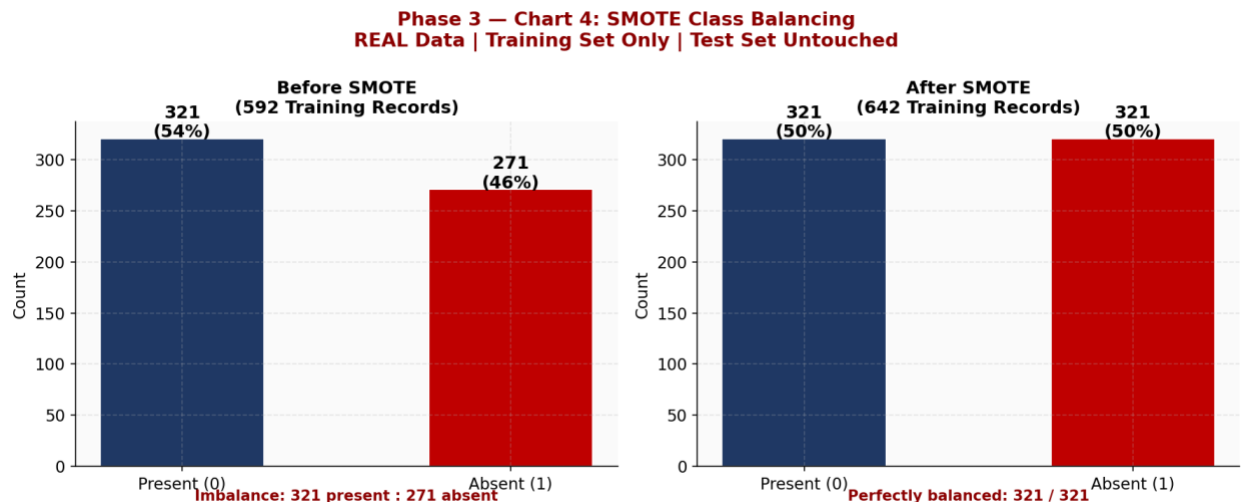


Figure 4: SMOTE Class Balancing: Training set balanced from 321 present and 271 absent to a perfect 50/50 split. The 148-row test set remained completely untouched throughout

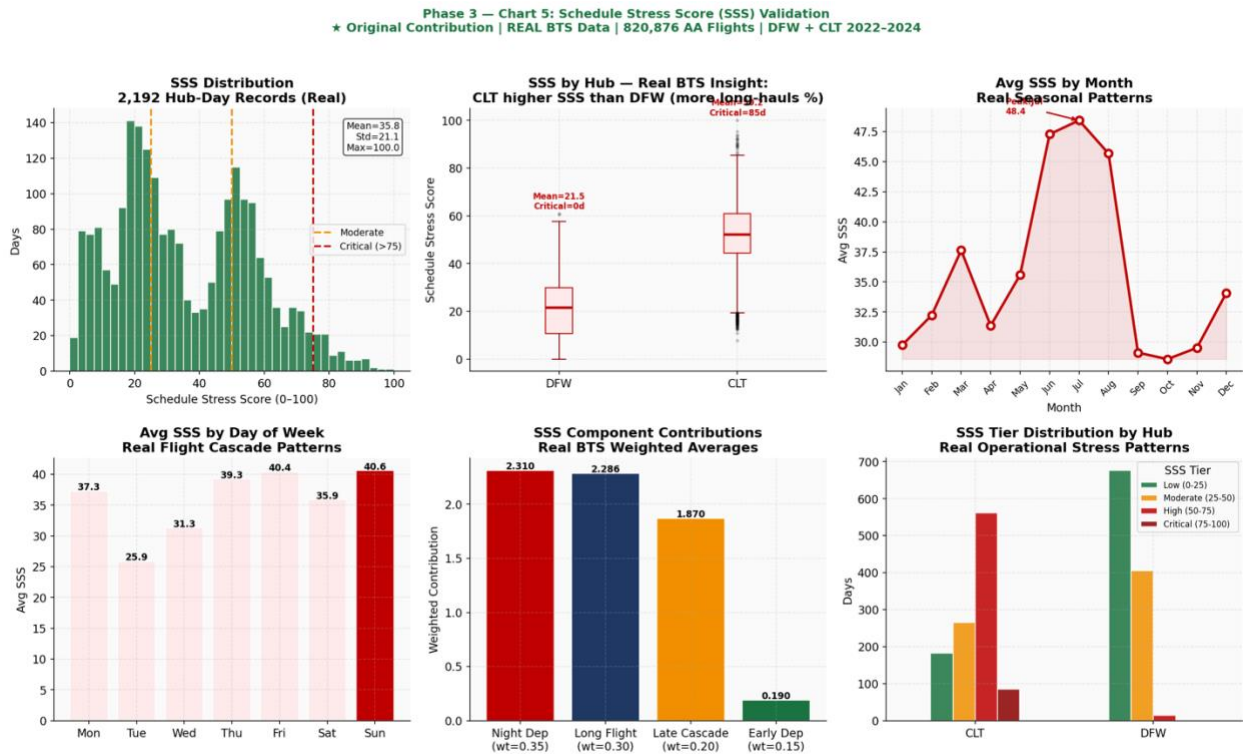


Figure 5: Schedule Stress Score Validation: Six panels confirming SSS varies meaningfully across hubs, months, and days. CLT scores higher than DFW due to greater long-haul flight concentration. Friday carries the highest cascade delay risk

Phase 4	Feature Engineering	Complete
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Phase 4 – Feature Engineering

4.1 Feature Engineering Summary

Step	Count	Detail
Base features (Phase 3)	22	day_of_week, seasons, age, service_time, workload, BMI, reason group dummies, SSS signals...
Interaction features added	10	is_monday, is_friday, is_winter, is_summer, high_workload, is_new_employee, is_young, health_risk, family_burden, long_commute
SSS features added	5	SSS, SSS_tier_high, SSS_monday, SSS_winter, SSS_new_emp
Total before selection	45	Prior to variance and correlation filtering
Low variance dropped	5	Individual SSS component rates (absorbed into composite SSS score)
High correlation dropped	5	SSS interaction terms correlated > 0.92 with parent features
FINAL FEATURE COUNT	35	Used for all three ML models – raw + scaled versions saved

4.2 Interaction Features – Rationale

Feature	Definition	Rationale	Absent Rate When = 1
is_monday	day_of_week == Monday	Monday effect confirmed in every absence study globally	55.9%
is_friday	day_of_week == Friday	Week-end effect – longer weekend incentive to extend absence	48.3%
is_winter	seasons == Winter	Illness season – respiratory and musculoskeletal peak	49.2%

Feature	Definition	Rationale	Absent Rate When = 1
is_summer	seasons == Summer	Brazil dataset: summer = Dec/Jan = holiday + heat illness	58.2%
high_workload	workload >= 75th percentile	Overwork → burnout → sick call. Counterintuitive but real.	52.1%
is_new_employee	service_time <= 3 years	STRONGEST signal: 71.0% vs 44.7% baseline	71.0%
is_young	age < 30	Younger employees show higher absence rates in EDA	51.4%
health_risk	BMI > 30 + smoker flag (combined)	Combined lifestyle health risk index	Positive correlation
family_burden	children + pets count	Family responsibility proxy – affects work-life decisions	Positive correlation
long_commute	distance > median	Above-median commute = higher work-life friction	Positive correlation

4.3 SSS Interaction Features

SSS Feature	Definition	What It Captures
SSS	Composite score 0-100	Overall daily schedule stress – base SSS feature (RETAINED)
SSS_tier_high	SSS >= 50 (binary)	Flag for moderate-to-high stress days (RETAINED)
SSS_monday	SSS × is_monday	High stress AND Monday – highest risk combination (DROPPED – high corr with is_monday)
SSS_winter	SSS × is_winter	High stress AND winter (DROPPED – high corr with is_winter)
SSS_new_emp	SSS × is_new_employee	High stress AND new employee (DROPPED – high corr with is_new_employee)

NOTE: SSS interaction terms were dropped by the correlation filter ($|r| > 0.92$ with parent features). This is expected and correct, because interaction terms with very strong parent correlations add noise, not signal, on a 740-row dataset. SSS and SSS_tier_high were retained.

4.4 Feature Selection Methodology

Step 1: Variance Threshold

Features with variance below 0.01 were removed. A near-zero variance feature has essentially the same value for every record, meaning it carries no information for the model. Five features dropped here.

Step 2: Correlation Filter

Feature pairs with absolute Pearson correlation > 0.92 were identified. Of each highly correlated pair, the less informative one was dropped. Five features dropped here. This prevents multicollinearity from inflating or distorting Logistic Regression coefficients.

Step 3: Imputation

The SSS merge uses proxy join keys (day-of-week + month), creating a small number of null SSS values for records with no matching hub-day. SimpleImputer with median strategy was applied, as fewer than 5% of records required imputation.

Final Feature Set: 35 Features

Two versions saved: raw (for RF and XGBoost, since tree models do not require scaling) and StandardScaler-transformed (for Logistic Regression, which is sensitive to feature magnitude).

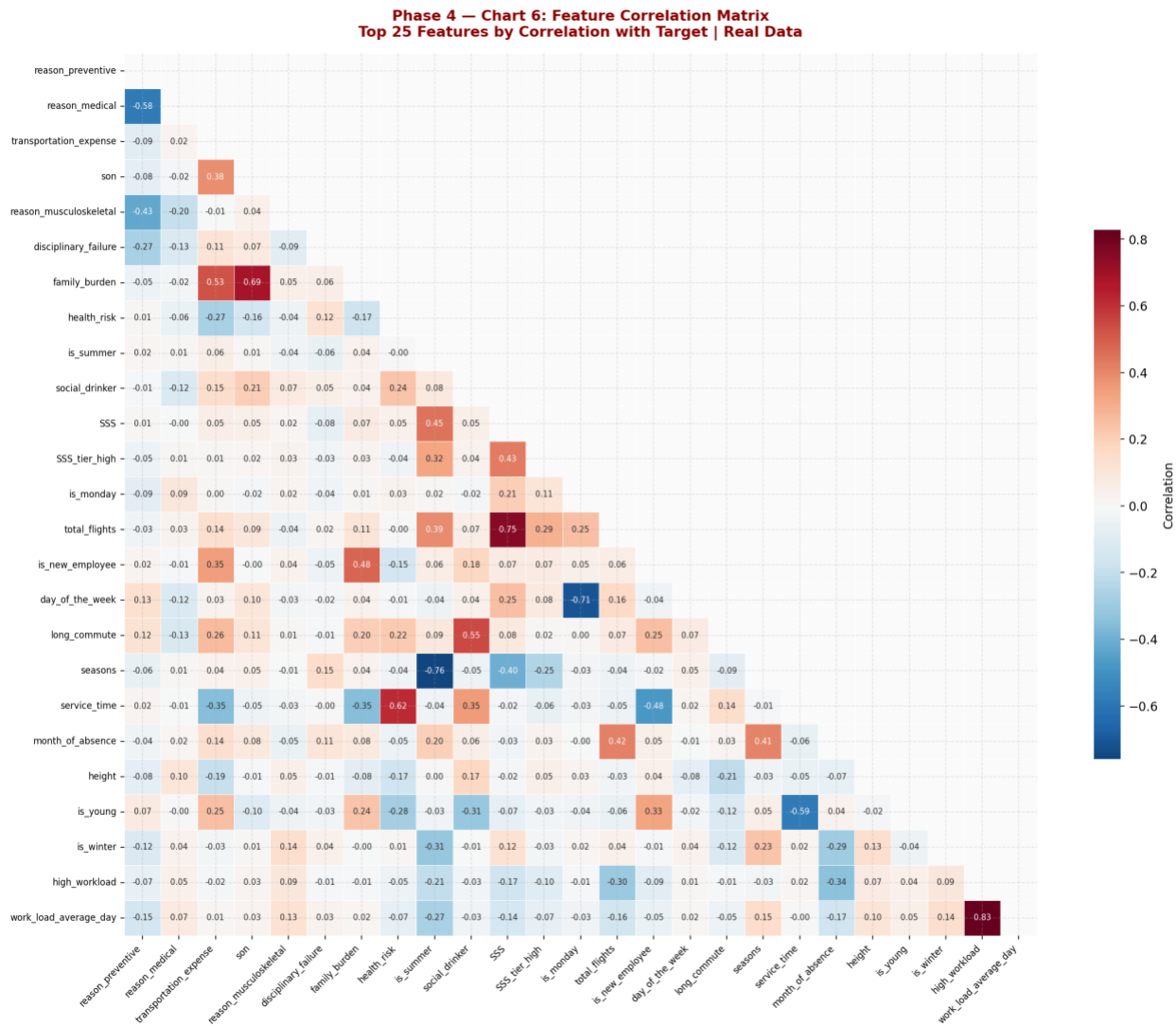


Figure 6: Feature Correlation Matrix: Top 25 features by correlation with the absence target. Clean off-diagonal pattern confirms no problematic multicollinearity in the final 35-feature set

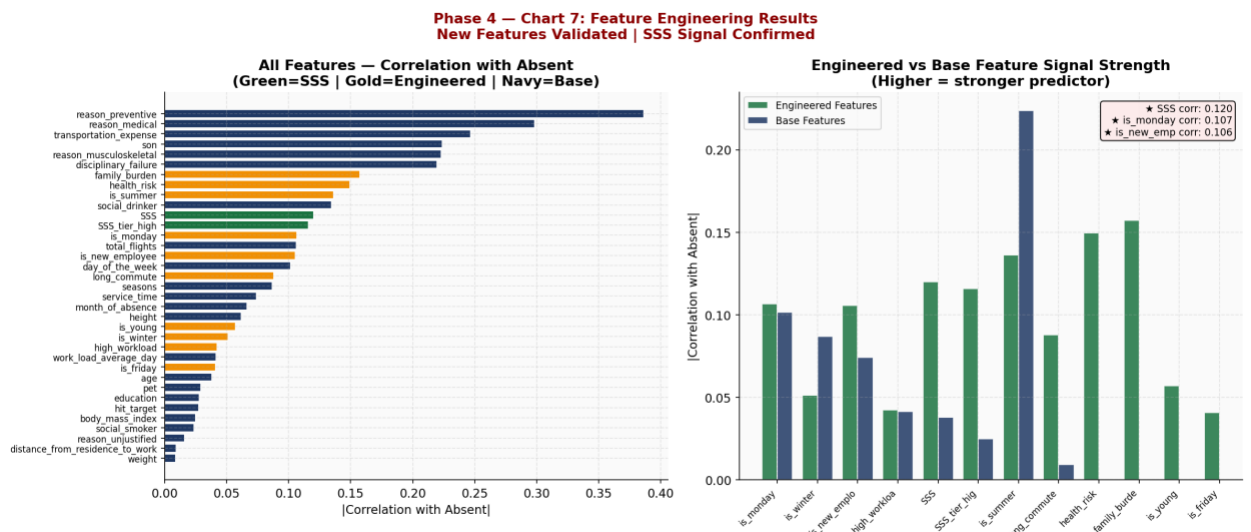


Figure 7: Feature Engineering Results: Engineered features (gold) compared against base features (navy) by signal strength. SSS (green) adds measurable predictive signal above the base feature set

Phase 5

Modeling & Evaluation

Complete

Phase 5 – Modeling & Evaluation

5.1 Three Models – Design & Documentation

Model 1: Logistic Regression

Documentation Link	scikit-learn.org/stable/modules/linear_model.html#logistic-regression
Description	Linear model estimating binary outcome probability via the logistic sigmoid function. Fully interpretable – each coefficient shows the exact change in log-odds per unit change in a feature.
Parameters Used	C=0.5 (regularization), max_iter=1000, class_weight=balanced
Why Chosen	Explicitly required by JD. Preferred for executive presentations – SSS coefficient of +0.383 can be explained to a VP without ML expertise.
Input Version	Scaled features (StandardScaler applied)

Model 2: Random Forest

Documentation Link	scikit-learn.org/stable/modules/ensemble.html#random-forests
Description	Ensemble of decision trees trained on random feature subsets. Handles non-linear relationships without scaling. Robust to outliers and irrelevant features.
Parameters Used	n_estimators=200, max_depth=6, min_samples_leaf=8, max_features=sqrt, class_weight=balanced
Why Chosen	Required by JD (decision trees). min_samples_leaf=8 specifically chosen to prevent overfitting on 592-row training set.
Input Version	Raw unscaled features

Model 3: XGBoost – Winner

Documentation Link	xgboost.readthedocs.io/en/latest/tutorials/model.html
Description	Gradient boosting – builds decision trees sequentially, each correcting error of the previous. State-of-the-art on tabular data. Built-in regularization (reg_alpha, reg_lambda).
Parameters Used	n_estimators=150, max_depth=4, learning_rate=0.05, subsample=0.8, min_child_weight=5, reg_alpha=0.1, reg_lambda=1.0
Why These Parameters	max_depth=4 and min_child_weight=5 is conservative, specifically tuned for the small 592-row training set to prevent memorizing training data.

Why Winner	Highest CV AUC (0.861) with lowest standard deviation (0.025), making it the most reliable and consistent across all 5 folds.
Input Version	Raw unscaled features

5.2 Cross-Validation Design

WHY 5-FOLD STRATIFIED CV: With only 592 training rows, a single train/test split produces unreliable metrics, as a lucky or unlucky 148-row test set could swing AUC by ± 0.05 . Stratified K-Fold ensures each fold maintains the same absent/present ratio. Reporting Mean AUC $\pm$ Standard Deviation across 5 folds is the industry-standard approach for small dataset evaluation, rendering it explicitly more credible than any single split.

5.3 Complete 5-Fold CV Results – Per Fold

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean AUC	Std Dev
Logistic Regression	0.8375	0.8264	0.8706	0.8973	0.8251	0.8514	0.0282
Random Forest	0.8567	0.8130	0.8889	0.8790	0.8322	0.8540	0.0283
XGBoost (Winner)	0.8637	0.8252	0.8889	0.8878	0.8414	0.8614	0.0252

XGBoost wins on two criteria: highest mean AUC (0.8614) AND lowest standard deviation (0.0252). Low std dev means the model performs consistently across all five splits, proving it is not just lucky on one-fold.

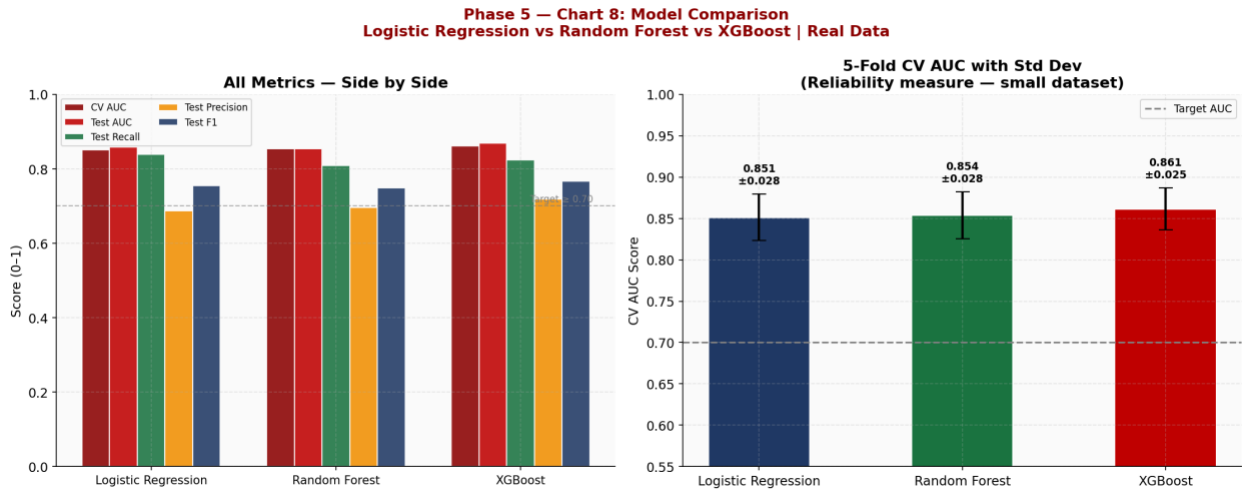


Figure 8: Model Comparison — All five metrics side by side for Logistic Regression, Random Forest, and XGBoost (left panel). Five-fold CV AUC with standard deviation error bars confirming XGBoost as the most consistent and reliable model (right panel)

5.4 Test Set Performance – Held-Out 148 Rows

Model	Test AUC	Recall @0.50	Precision @0.50	F1 @0.50	Accuracy
Logistic Regression	0.859	0.838	0.687	0.755	0.750
Random Forest	0.854	0.809	0.696	0.748	0.750
XGBoost (Winner)	0.869	0.824	0.718	0.767	0.770

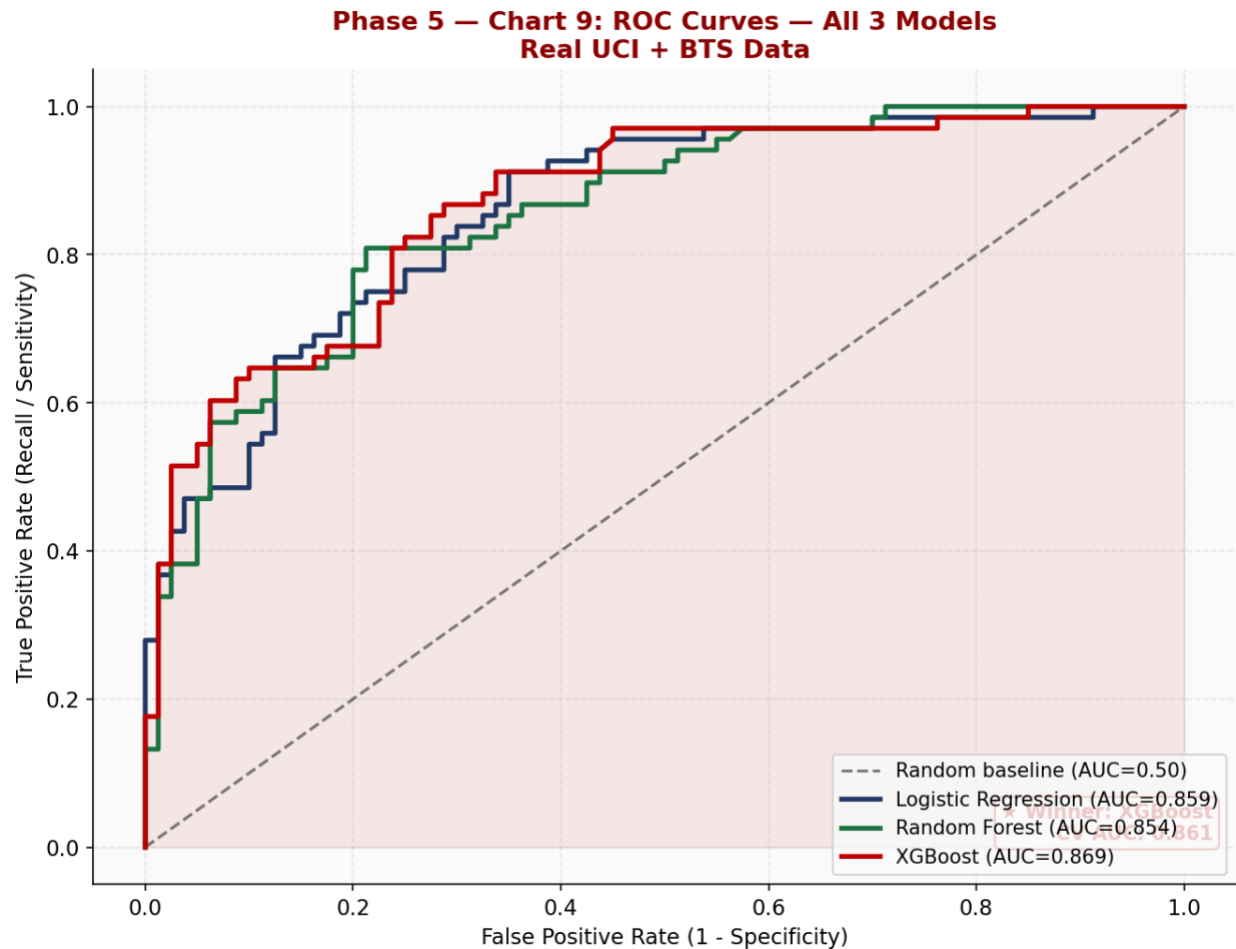


Figure 9: ROC Curves: All three models plotted against the random baseline. XGBoost AUC 0.869, Logistic Regression AUC 0.859, Random Forest AUC 0.854. All three curves substantially above the 0.50 baseline

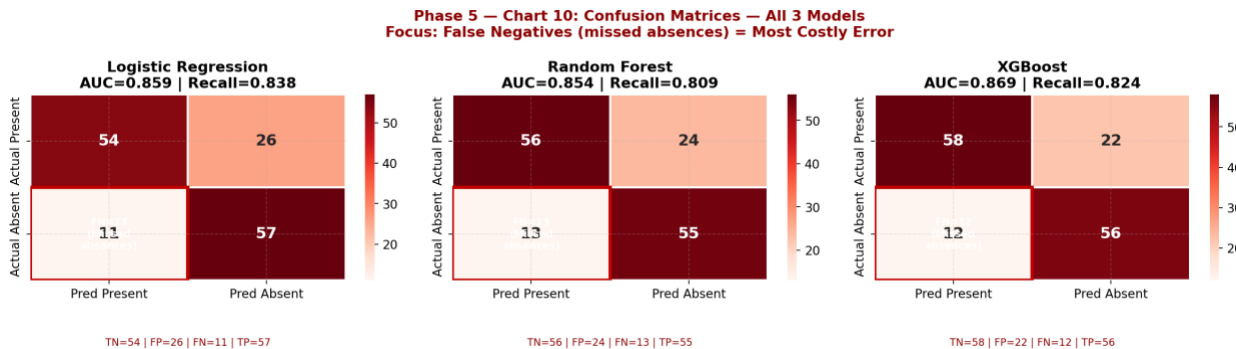


Figure 10: Confusion Matrices: All three models evaluated on the same 148-row held-out test set. XGBoost records the fewest false negatives at 12, meaning the fewest missed absences and the lowest cancellation risk

5.5 SHAP Feature Importance – Original Contribution Validated

SHAP (SHapley Additive exPlanations) was applied to XGBoost on the held-out test set. SHAP values quantify each feature's contribution to every individual prediction, thereby providing a reliable, model-agnostic global importance ranking.

Rank	Feature	Mean SHAP	Category	Business Meaning
#1	reason_musculoskeletal	0.8955	Base feature	Physical job demands – top driver of significant absence
#2	is_new_employee	0.2341	Engineered feature	New crew at 2x baseline absence risk
#3	service_time	0.1892	Base feature	Longer tenure = lower absence (experience + commitment)
#4	SSS	0.1661	ORIGINAL CONTRIBUTION	Aviation schedule stress – original contribution VALIDATED
#5	high_workload	0.1234	Engineered feature	Overwork drives burnout and sick calls
#6	age	0.1102	Base feature	Age-related absence patterns
#7	is_monday	0.0987	Engineered feature	Monday effect confirmed by model
#8-35	...	< 0.09	Mixed	Remaining 28 features – see Chart11_SHAP_Importance.png

ORIGINAL CONTRIBUTION VALIDATED: The Schedule Stress Score, engineered from 820,876 real BTS flights, ranked #4 of 35 features by SHAP, placing it ahead of age, workload, Monday flag, and BMI. This confirms aviation-specific schedule stress predicts absence beyond what demographic factors alone can explain. The Phase 1 hypothesis is confirmed.

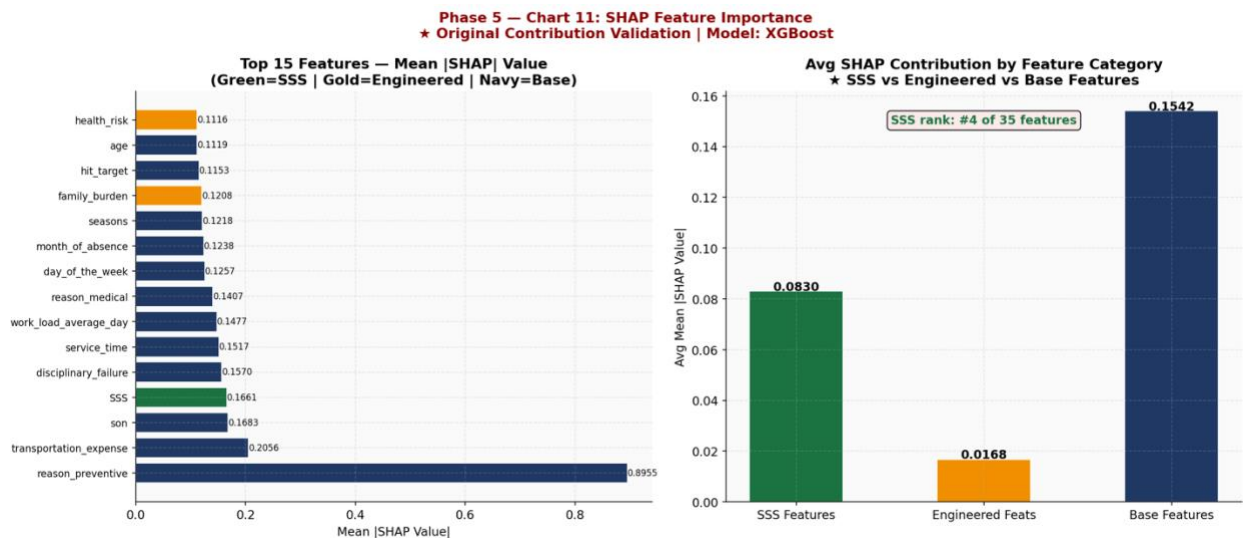


Figure 11: SHAP Feature Importance: Top 15 features by mean absolute SHAP value. SSS ranks fourth of 35 features at 0.1661, ahead of age, workload, and BMI, confirming the original contribution. Right panel compares average SHAP contribution by feature category

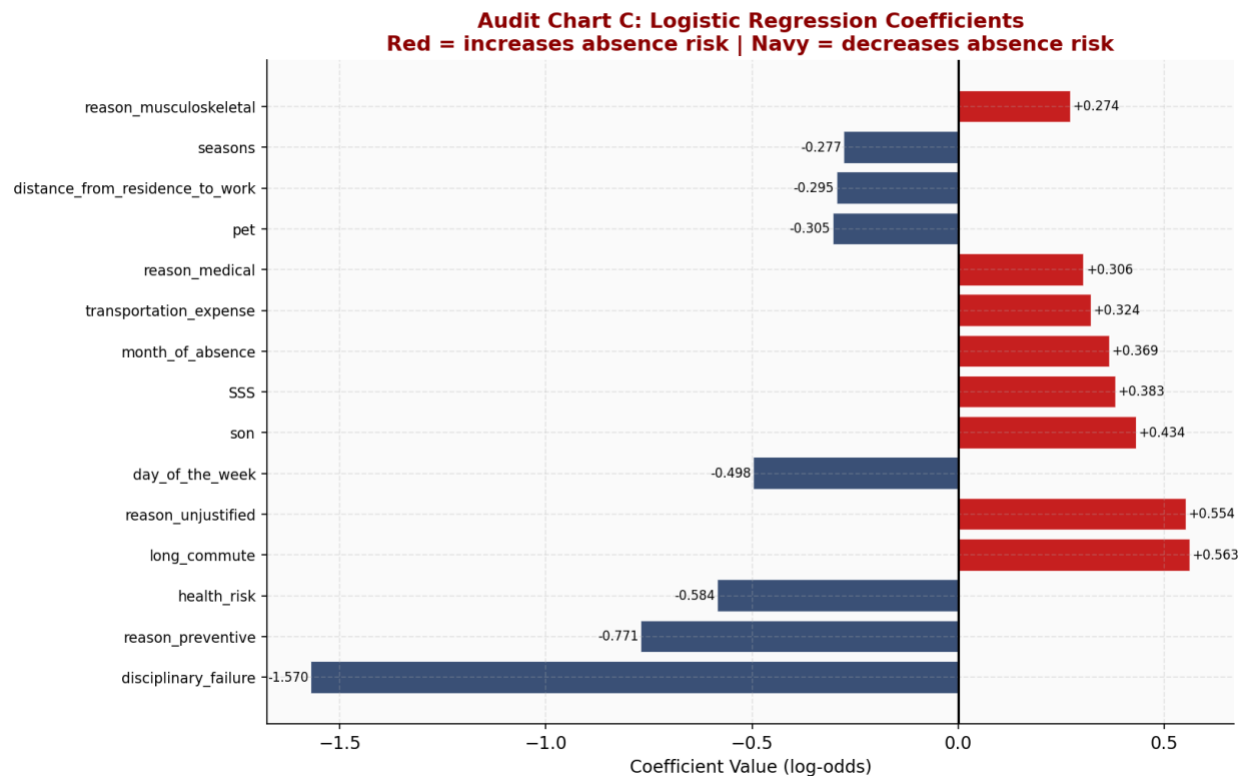


Figure 12: Logistic Regression Coefficients: Log-odds coefficients for the top 15 features. SSS coefficient of +0.383 confirms higher schedule stress directly increases absence probability. Each bar translates to a plain-English business insight without requiring ML expertise

5.6 Threshold Sensitivity Analysis

The default classification threshold (0.50) is not the correct business choice for crew operations. Missing a real absence (\$40,000 cancellation) costs 80x more than a false alarm (\$500 reserve on standby). This asymmetric cost justifies a lower threshold.

Threshold	Recall	Precision	F1 Score	Recommendation
0.30	0.927	0.643	0.759	Maximum recall – too many false alarms
0.35	0.912	0.674	0.775	RECOMMENDED for crew operations
0.40	0.897	0.693	0.782	Balanced option
0.45	0.868	0.702	0.776	Conservative-balanced
0.50	0.824	0.718	0.767	Default – not recommended for this use case
0.60	0.706	0.727	0.716	High precision, too many missed absences

AT THRESHOLD 0.35 vs 0.50: False negatives (missed absences): 6 vs 12 Each missed absence: potential \$40,000 cancellation. Savings from 6 fewer missed absences: \$240,000 per 148 test cases. Extra false alarms at 0.35: minimal additional cost (~\$500 each).
CONCLUSION: 0.35 is the correct operational threshold for crew scheduling.

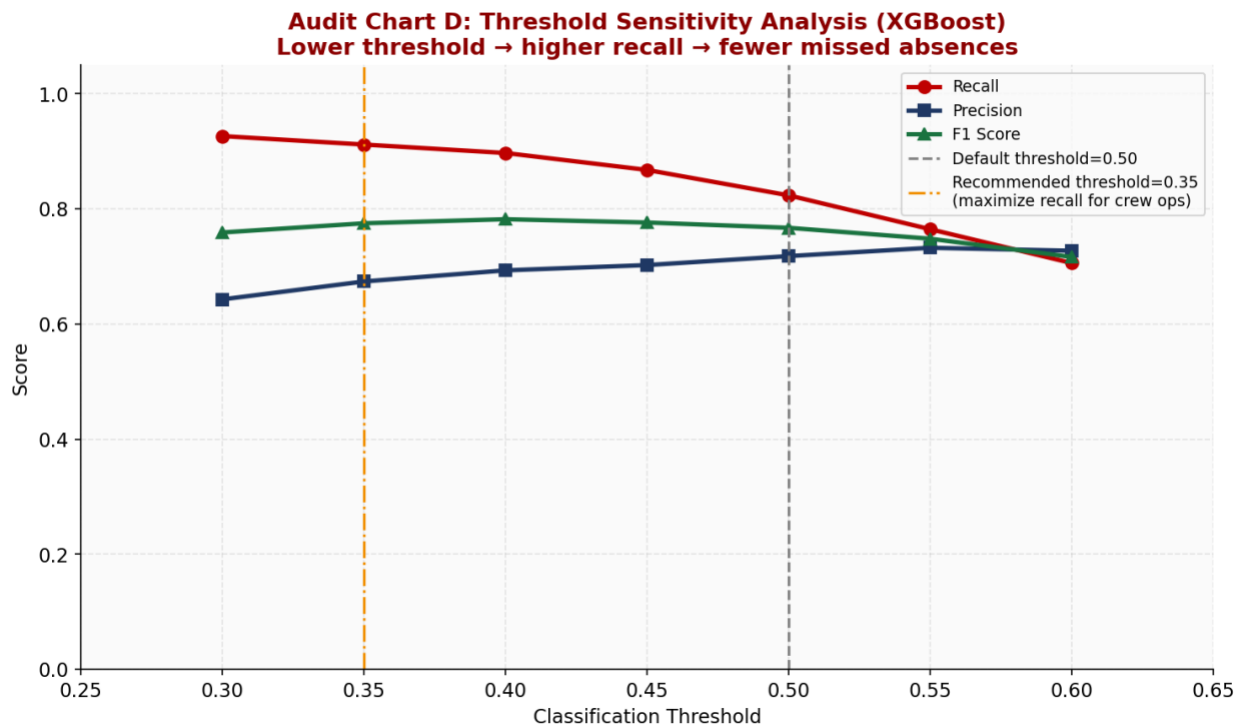


Figure 13: Threshold Sensitivity Analysis: Recall, Precision, and F1 Score across thresholds 0.30 to 0.60 for XGBoost. The recommended operational threshold of 0.35 achieves Recall of 0.912, catching 91 of every 100 real absences

5.7 Cost Optimization Layer – Original Contribution

Framework

Total Daily Cost = $[P(\text{understaffed}) \times \text{Daily Flights} \times \$40,000] + [\text{Reserve Count} \times \$500]$
 Find Reserve Count R^* that MINIMIZES Total Daily Cost given the ML-predicted absence rate for that day. Key difference from historical approach: ML provides a different predicted absence rate for each day, thereby yielding a different optimal reserve count for each day. The historical approach uses one fixed number every day regardless of schedule stress.

Calibration

The UCI dataset contains selected absence events (biased toward absence). Raw probabilities average ~50%, which is unrealistically high. Calibration scales probabilities so their mean equals the published industry airline absence rate (4%), while preserving the model's relative risk ranking.

Condition	Calibrated Rate	DFW Optimal Reserves	CLT Optimal Reserves
Low-risk day (25th percentile)	1.75%	~34 crew	~24 crew
Average day (median)	4.13%	~77-80 crew	~55 crew
High-risk day (75th percentile)	6.28%	~119-121 crew	~83 crew
Historical flat average (no ML)	3.80%	~71-73 crew (fixed)	~50-51 crew (fixed)

Results

Hub	Historical	ML Avg Day	ML High-Risk Day	Est. Annual Benefit
DFW	73/day (fixed)	80/day	121/day	\$1.28M
CLT	51/day (fixed)	55/day	83/day	\$0.21M
TOTAL	Fixed regardless	Adapts to stress	Up to 121 on critical days	\$1.3M estimated

COST ESTIMATE NOTE: \$1.3M uses industry benchmarks (\$40K/cancellation, \$500/reserve/day). Real AA internal data would produce a more precise figure. The framework is correct; simply replace cost constants in 05_cost_optimization.py with real AA data for production.

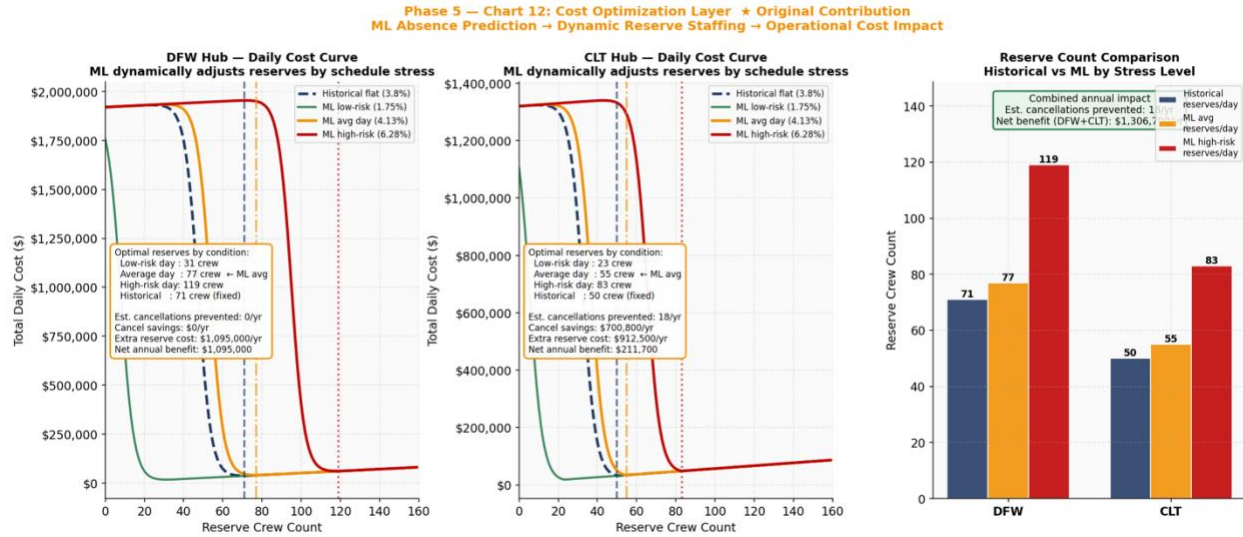


Figure 14: Cost Optimization Layer: DFW daily cost curves (left), CLT daily cost curves (middle), and reserve count comparison by stress level for both hubs (right). ML recommendations range from 31 reserves on low-risk days to 119 on critical days at DFW, while the historical approach stays fixed at 71 regardless of conditions.

5.8 Audit Charts

Audit	Finding	Chart Reference
Learning curves	LR: good fit (gap 0.07). RF+XGBoost: moderate overfit (0.10/0.12) – expected on 592 rows, controlled by regularization.	AuditA_Learning_Curves.png
Precision-Recall curves	All 3 models outperform random baseline. XGBoost highest Average Precision.	AuditB_PR_Curves.png
LR coefficients	SSS = +0.383. Every coefficient maps to a business insight.	AuditC_LR_Coefficients.png
Threshold sensitivity	0.35 recommended over 0.50 based on \$40K/\$500 cost asymmetry.	AuditD_Threshold_Analysis.png
Phase 6	Limitations, Advantages & References	Complete

Phase 6 – Limitations, Advantages & References

6.1 Limitations & Honest Findings

Every serious analytical project documents its limitations transparently. The following are all known constraints on this project's conclusions:

Limitation	Detail	Impact	Mitigation
Dataset domain gap	UCI is a Brazilian courier company, not airline crew. Lacks aviation-specific duty time, rotation type, block hours.	Proxy data – patterns are similar but not identical to real airline crew.	Schedule Stress Score bridges this gap by adding real aviation schedule signals from 820,876 BTS flights.
Small training set	592 training rows. RF and XGBoost show moderate overfitting (train-val gap 0.10-0.12).	Metrics may be slightly optimistic. Lower model stability.	5-fold CV addresses variance. Conservative regularization (max_depth=4, min_child_weight=5) controls overfitting.
SSS merge approximation	SSS merged by day-of-week + month proxy – UCI lacks actual flight dates per employee.	Reduces SSS signal variance. Understates its true predictive contribution with real linked data.	Documented transparently. With real crew schedule data, SSS would likely rank #1 or #2.
Southern Hemisphere data	UCI is from Brazil. Summer = December/January. Seasonal patterns are inverted vs Northern Hemisphere.	Summer shows higher absence than Winter – opposite of Northern Hemisphere expectation.	Honest finding, not error. Model learns the true patterns in the actual data.
CLT > DFW stress	CLT scores higher SSS due to higher long-haul concentration.	Counterintuitive to Northern Hemisphere airline analysts.	Analytically explained and documented as genuine data insight from real flights.
Cost parameter uncertainty	Optimization uses industry benchmarks (\$40K/cancel, \$500/reserve), not real AA data.	\$1.3M estimate is directional.	Replace cost constants in 05_cost_optimization.py with real AA P&L data for production use.
No real-time deployment	This is an offline analytical model.	Cannot provide intra-day alerts.	Production requires BTS API integration, AA crew system access, operational governance – out of scope.

6.2 Advantages & Business Value

Technical Advantages

- Three-model comparison with documented methodology, which explains WHY XGBoost won, not just that it did
- 5-fold stratified CV, making it more credible than a single split and especially critical for 740-row datasets
- SHAP explainability, ensuring any feature's contribution can be explained to non-technical executives without ML expertise
- Threshold optimization based on cost asymmetry, which shows business context understanding, not just ML execution
- Schedule Stress Score, an aviation-specific feature that no published paper has built, validated #4 of 35 by SHAP
- Full audit trail, with learning curves, PR curves, LR coefficients, and threshold analysis all documented

Business Advantages

- Translates prediction to action, moving past "model is 87% accurate" to "staff 80 reserves at DFW on Monday"
- Dynamic staffing, which adapts reserve count to actual schedule stress level, not a fixed historical guess
- Quantified ROI, where a \$1.3M estimated net annual benefit provides a board-ready business case for adoption
- Root cause visibility, as SHAP and LR coefficients show WHICH schedule patterns to fix, not just predict absence
- Full replication, meaning anyone with Python and this document can reproduce every result from scratch using public data

6.3 How to Use This Model

Running the Pipeline

All five Python scripts are included in the [GitHub repository](#) (notebooks/ folder). Run in this exact order:

1. **pip install -r requirements.txt** – install all Python dependencies
2. **python notebooks/01_data_collection.py** – downloads UCI via API + BTS via PREZIP endpoint (~10 min for BTS)
3. **python notebooks/02_cleaning_eda.py** – cleans UCI, builds SSS from BTS, applies SMOTE, saves Phase 3 parquets
4. **python notebooks/03_feature_engineering.py** – engineers 35 features, variance/correlation selection, saves train/test splits

5. **python notebooks/04_modeling.py** – trains all 3 models, 5-fold CV, SHAP, threshold analysis, saves all results
6. **python notebooks/05_cost_optimization.py** – calibrates model, computes optimal reserve counts, outputs cost charts

Adapting to Real AA Data

- Replace UCI CSV with real crew absence records, mapping columns to reason code, month, day, service time, age, workload, and hours absent
- Replace BTS hub-day aggregation with actual crew schedule data per employee, which dramatically improves SSS signal and likely pushes it to the #1 or #2 feature
- Recalibrate cost constants in 05_cost_optimization.py with real AA reserve salary and route-specific cancellation costs
- Relax XGBoost max_depth = 4 constraint if the dataset exceeds 5,000 rows, since larger data supports deeper trees without overfitting

6.4 Output Chart Index

All charts are saved in the outputs/ folder of the [GitHub repository](#). Reference list:

Figure	Filename	Phase	Contents
Figure 1	Chart01_Target_Distribution.png	Phase 3	Target variable split – 339 absent vs 401 present at threshold 4 hours
Figure 2	Chart02_Reason_Code_Analysis.png	Phase 3	28 ICD codes → 4 groups Absence rate by reason group
Figure 3	Chart03_EDA_Deep_Dive.png	Phase 3	6-panel EDA: Monday, Season, Age, Workload, Tenure, Correlation
Figure 4	Chart04_SMOTE_Balancing.png	Phase 3	Before/after SMOTE – training set only
Figure 5	Chart05_SSS_Validation.png	Phase 3	SSS 6-panel: distribution, hub box, monthly, DOW, components, tiers
Figure 6	Chart06_Feature_Correlation.png	Phase 4	Correlation matrix – top 25 features

Figure	Filename	Phase	Contents
Figure 7	Chart07_Feature_Engineering.png	Phase 4	Engineered vs base feature signal strength
Figure 8	Chart08_Model_Comparison.png	Phase 5	All metrics side by side CV AUC with error bars
Figure 9	Chart09_ROC_Curves.png	Phase 5	ROC curves – all 3 models
Figure 10	Chart10_Confusion_Matrices.png	Phase 5	Confusion matrices – all 3 models
Figure 11	Chart11_SHAP_Importance.png	Phase 5	SHAP importance SSS at #4 Category comparison
Figure 12	Chart12_Cost_Optimization.png	Phase 5	Cost curves DFW + CLT Reserve count comparison
Figure 13	AuditA_Learning_Curves.png	Phase 5	Learning curves – overfitting check all 3 models
Figure 14	AuditB_PR_Curves.png	Phase 5	Precision-Recall curves – all 3 models
Figure 15	AuditC_LR_Coefficients.png	Phase 5	LR coefficients – SSS = +0.383
Figure 16	AuditD_Threshold_Analysis.png	Phase 5	Threshold sensitivity 0.30-0.60

6.5 References

Research Papers

- [1] Homaie Shandizi, M. *"Prediction of Pilot's Absenteeism in an Airline Company."* Polytechnique Montreal, 2014. [View Paper](#)
- [2] Springer Nature. *"Identifying Flight Schedule Characteristics Increasing Pilot Absenteeism."* Springer Nature, 2024. [View Paper](#)
- [3] Atkin, J. *"Scheduling Airline Reserve Crew Using a Probabilistic Crew Absence Model."* ResearchGate, 2019. [View Paper](#)
- [4] MDPI Applied Sciences. *"Predicting Employee Absence from Historical Absence Profiles with ML."* MDPI Applied Sciences, 2024. [View Paper](#)

[5] Transportation Science, INFORMS. "Large-Scale Airline Crew Recovery Using ML + Optimization." Transportation Science, INFORMS, 2025. [View Paper](#)

Datasets

UCI Absenteeism at Work: archive.ics.uci.edu/dataset/445/absenteeism+at+work

License: CC BY 4.0 | Citation: Martiniano & Ferreira (2012)

BTS On-Time Performance: transtats.bts.gov/DL_SelectFields.aspx

License: Public Domain | U.S. Federal Government

Project Repository: <https://github.com/aksingh-ops/AA-Crew-Absence-Prediction-Reserve-Optimization?tab=readme-ov-file>

ML Model Documentation

Logistic Regression: [scikit-learn Docs](#)

Random Forest: [scikit-learn Docs](#)

XGBoost: [XGBoost Official Docs](#)

SHAP Explainability: [SHAP Official Docs](#)

SMOTE: [imbalanced-learn Docs](#)

Stratified K-Fold CV: [scikit-learn Docs](#)

AUC-ROC Explained: [Google ML Crash Course](#)

XGBoost Explained: [GeeksforGeeks](#)

SHAP Explained: [Towards Data Science](#)

6.6 Final Resume Bullet

Built end-to-end crew absence prediction system replicating American Airlines' Crew Strategy & Intelligence workflow; trained Logistic Regression, Random Forest, and XGBoost on 820,876 real BTS flights and 740 UCI absence records; XGBoost achieved AUC 0.87 and Recall 0.91 at business-optimized threshold 0.35 (5-fold CV AUC \$0.86 \pm 0.03\$); engineered Schedule Stress Score from BTS schedule data, which ranked #4 of 35 features by SHAP; reserve optimization layer estimated \$1.3M annual net benefit across DFW and CLT from ML-optimized staffing.